Practice Test B — Answer Key & Worked Solutions

One worked solution per question. Mark the working as well as the final answer.

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1. [Solving Linear Equations (for x)]

Answer: C) $x = 3$

Worked solution: Step 1: collect x-terms on one side and constants on the other → $5x - 2x = 6 + 3$. Step 2: combine → $3x = 9$. Step 3: divide by 3 → $x = 3$. Check: $5(3) - 3 = 12$ and $2(3) + 6 = 12$ ✓.

2. [Literal Equations & Solving for y]

Answer: A) $r = C \div (2\pi)$

Worked solution: Step 1: divide both sides by 2π. Step 2: $r = C \div (2\pi)$.

3. [Linear Inequalities]

Answer: A) $x < -4$

Worked solution: Step 1: divide both sides by -2. Step 2: dividing by a negative flips the sign → $x < -4$.

4. [Systems of Linear Equations]

Answer: A) (5, 2)

Worked solution: Elimination: Step 1: add the equations → $(x + x) + (y - y) = 7 + 3 \rightarrow 2x = 10$. Step 2: $x = 5$. Step 3: plug back → $5 + y = 7 \rightarrow y = 2$. Solution: (5, 2).

5. [Functions: Notation, Domain & Range]

Answer: B) 7

Worked solution: Step 1: substitute 3 for x → $g(3) = 3^2 - 2$. Step 2: square → $9 - 2$. Step 3: subtract → 7.

6. [Linear Functions: Slope, Intercepts & Equations]

Answer: A) (0, 1)

Worked solution: The y-intercept is where $x = 0$: $y = 2(0) + 1 = 1$, so the line crosses the y-axis at (0, 1).

7. [Quadratic Functions & Equations]

Answer: 3

Worked solution: Step 1: factor → $(x - 3)(x + 3) = 0$. Step 2: $x = 3$ or $x = -3$. The positive solution is 3.

8. [Polynomials: Operations & Factoring]

Answer: A) $x^2 + 5x + 6$

Worked solution: FOIL: First $x \cdot x = x^2$; Outer $x \cdot 3 = 3x$; Inner $2 \cdot x = 2x$; Last $2 \cdot 3 = 6$. Combine: $x^2 + 5x + 6$.

9. [Radical & Rational Expressions]

Answer: A) $x = 12$

Worked solution: Step 1: multiply both sides by 2 → $x = 6 \times 2$. Step 2: $x = 12$.

10. [Exponential Functions & Growth/Decay]

Answer: A) exponential decay

Worked solution: Step 1: the base $b = 0.8$ is between 0 and 1. Step 2: $0 < b < 1$ means decay.

11. [Data: Measures of Center & Spread]

Answer: 4

Worked solution: Step 1: count how often each value appears → 4 appears twice, all others once. Step 2: the mode is 4.

12. [Scatter Plots, Line of Best Fit & Correlation]

Answer: A) approximates the overall trend of the data

Worked solution: The line of best fit summarizes the general direction of the points; it does not need to touch every point.

13. [Absolute Value Equations & Inequalities]

Answer: B) $x = 8$ or $x = -2$

Worked solution: Split into two cases. Case 1: $x - 3 = 5 \rightarrow x = 8$. Case 2: $x - 3 = -5 \rightarrow x = -2$. Check: $|8 - 3| = 5$ ✓ and $|-2 - 3| = |-5| = 5$ ✓.

14. [Arithmetic & Geometric Sequences]

Answer: 79

Worked solution: $a_1 = 3$ and $d = 4$. $a_n = a_1 + (n - 1)d$ gives $a_{20} = 3 + 19(4) = 3 + 76 = 79$.

15. [Inverse Functions]

Answer: 2

Worked solution: $f^{-1}(0)$ asks which input gives an output of 0. Solve $3x - 6 = 0 \rightarrow 3x = 6 \rightarrow x = 2$. Check: $f(2) = 3(2) - 6 = 0$ ✓.

16. [Function Transformations & Vertex Form]

Answer: A) shifted right 2 units

Worked solution: Changes inside the function act on the input and run opposite to the sign. $f(x - 2)$ needs an input 2 larger to produce the same output, so the graph moves right 2 units.

17. [Systems of Linear Inequalities & Linear Programming]

Answer: D) (1, 1)

Worked solution: Check the sum first, then the signs. (1, 1): $1 + 1 = 2 \leq 4$ ✓, both coordinates non-negative ✓. (3, 2) gives 5, which is more than 4. $(-1, 2)$ fails $x \geq 0$. (4, 1) gives 5, which is more than 4.

18. [Probability from Two-Way Tables]

Answer: 0.25

Worked solution: The “given walks” shrinks the sample space to the 120 walkers. Of those, 30 are late, so $P = 30/120 = 0.25$.

19. [Quadratic Functions & Equations]

Answer: A) $x = -4$ or 2

Worked solution: Step 1: factor $\rightarrow (x + 4)(x - 2) = 0$. Step 2: $x + 4 = 0 \rightarrow x = -4$; $x - 2 = 0 \rightarrow x = 2$.

20. [Polynomials: Operations & Factoring]

Answer: A) $(x + 4)(x - 4)$

Worked solution: Step 1: recognize a difference of squares $a^2 - b^2$ with $a = x$ and $b = 4$. Step 2: $x^2 - 16 = (x + 4)(x - 4)$.

21. [Radical & Rational Expressions]

Answer: 9

Worked solution: Step 1: subtract 1 from both sides $\rightarrow x/3 = 3$. Step 2: multiply by $3 \rightarrow x = 9$.

22. [Exponential Functions & Growth/Decay]

Answer: A) exponential growth

Worked solution: Step 1: the base $b = 1.5$ is greater than 1. Step 2: $b > 1$ means growth.

23. [Data: Measures of Center & Spread]

Answer: 7

Worked solution: Step 1: find the max = 9 and the min = 2. Step 2: range = max − min = 9 − 2 = 7.

24. [Scatter Plots, Line of Best Fit & Correlation]

Answer: A) weak or no linear correlation

Worked solution: r ranges from −1 to 1; values near 0 mean there is little to no linear relationship.

Marking your test

- One mark per question, 24 marks in total. Give the mark only if the working supports the answer.
- 20 or more: the topics are solid — move on to timed mixed practice.
- 15 to 19: reread the method for each topic you dropped a mark in, then redo those questions in the app.
- 14 or fewer: work through the topic list in the app one topic at a time, then take the next practice test.
- Re-take any test after a week. If your score has not moved, that topic needs a different explanation, not more repetition.
